

ELI — Architecture Ascii

ELI v2.3.72

September 2026

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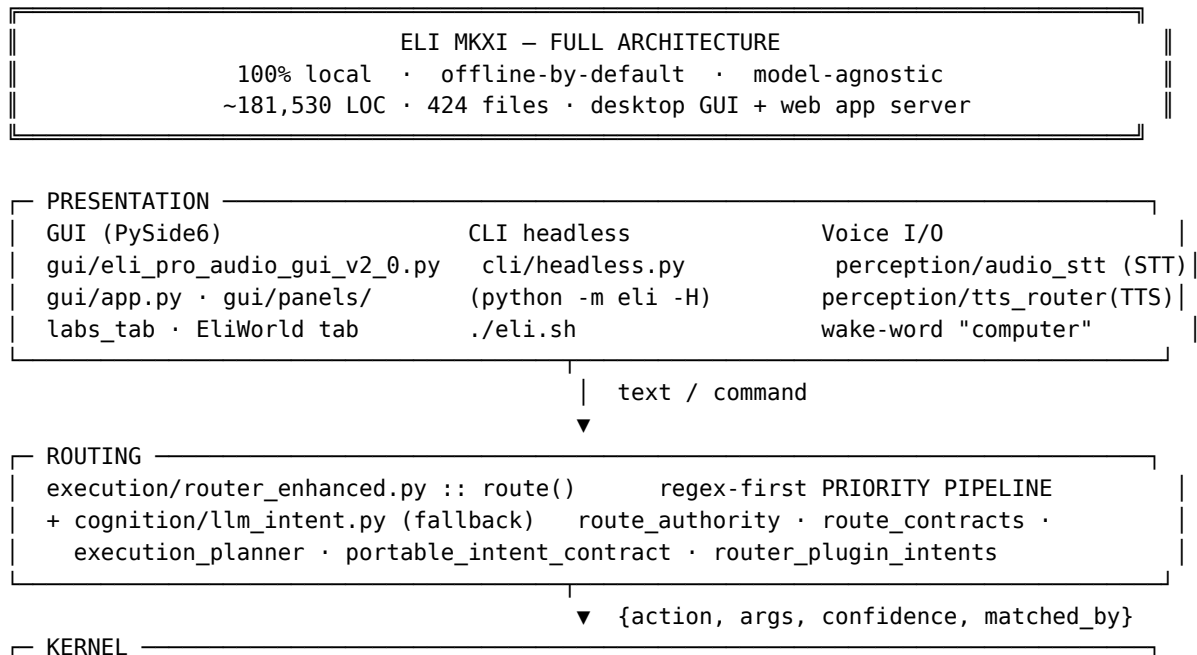
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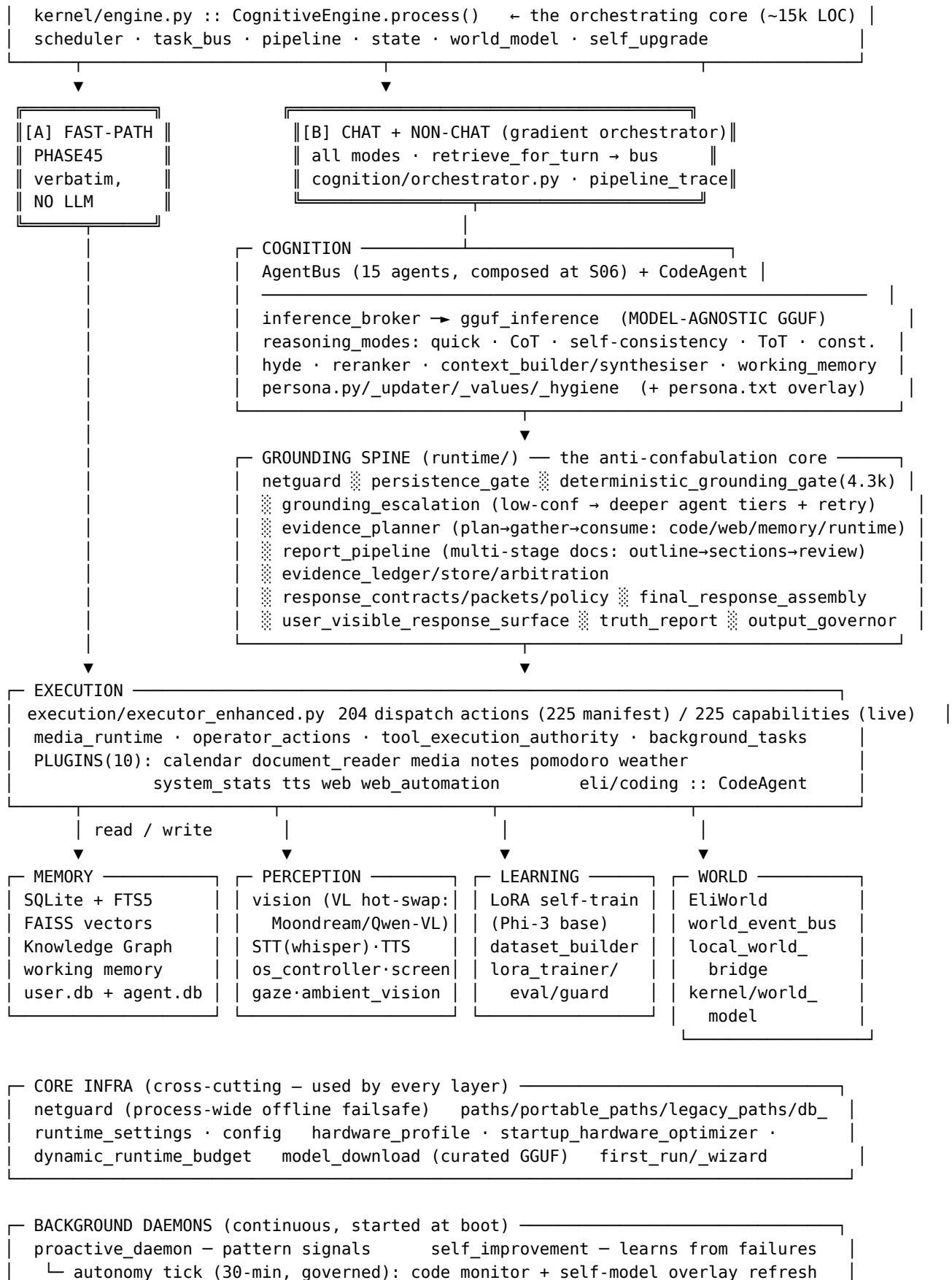
Blueprint – ELI MKXI Full Architecture (ASCII)

Updated for v2.3.72. All CHAT modes → gradient orchestrator; bus composed at S06.

The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see `architecture.md` for prose, `diagrams.md` for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

A. Full system — layered block diagram





+ goal/scheduler ticks → proposals (observe-only / memory-write; need approval)
habits_scheduler/habits – routines scheduler · task_bus – jobs
background_tasks – async heavy work reflection loop ambient_vision loop
scheduled_tasks (durable overnight/timed) world_event_bus ← confidence events

OUTPUT ← output_governor / sanitiser ← inference ← (any path)

└→ TTS (Piper) / GUI render

B. The 15 agents

AGENT BUS (cognition/agent_bus.py :: AgentBus.dispatch → DispatchResult)
selects a set → runs on a dependency DAG → aggregates grounding_confidence

BusMemoryAgent	recall: SQLite/FTS5/FAISS → context
KnowledgeGraphAgent	entities/relations from the KG
SystemAgent	runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)
OrchestratorAgent	plan / coordinate grounded synthesis
FileCodeAgent	searches the codebase for code-grounded answers
IntrospectionBus...	live runtime + gathers identity/awareness audits as evidence (persona summarises – never a data dump)
CapabilityAgent	capability manifest (what ELI can do)
ReflectionAgent	session reflection / insights
ProactiveAgent	proactive patterns / signals
HabitAgent	learned habits
SelfImprovementAgent	self-improvement state
FrontierAgent	frontier reasoning hooks
PluginAgent	dispatch to registered plugins
VoiceAgent	STT/TTS engine state
CriticAgent	second-tier verifier: checks retriever agents' output
+ CodeAgent (16th)	SEPARATE pipeline: plan→search→verify→repair
eli/coding/agent	(CODE_SOLVE / GENERATE_SCRIPT) – not a bus agent

C. Module tree (LOC · key files · role)

eli/ (~181,530 LOC, 424 files) · api/server.py (FastAPI web app + dashboard)

```

├── __main__.py ..... entry dispatch (GUI | --headless)
├── kernel/          16.7k — the core
│   ├── engine.py    15.2k   CognitiveEngine.process() – the spine *god-file
│   ├── scheduler.py          timed jobs
│   └── task_bus.py · pipeline.py · state.py · world_model.py · self_upgrade.py
├── execution/       26.2k — route → act
│   ├── executor_enhanced.py  15.9k   204 dispatch / 225 manifest *god-file
│   ├── router_enhanced.py    8.2k    priority pipeline *god-file
│   ├── execution_planner.py · route_authority.py · route_contracts.py
│   └── tool_execution_authority.py · operator_actions.py · operator_policy.py

```

- └─ media_runtime.py · router_plugin_intents.py · executor_plugin_handlers.py
- └─ portable_intent_contract.py · execution_intent_packets.py
- cognition/ 18.5k — think
 - └─ agent_bus.py 2.4k 15 agents + dispatch *
 - └─ orchestrator.py 12-stage deep retrieval
 - └─ gguf_inference.py 2.1k · inference_broker.py model-agnostic inference
 - └─ reasoning_modes.py · hyde.py · reranker.py · llm_intent.py
 - └─ persona.py/_updater/_values/_status/_hygiene (+ persona.txt, persona.auto.txt)
 - └─ context_builder.py · context_synthesiser.py · working_memory.py
 - └─ output_governor.py · response_governance.py · response_sanitizer.py
 - └─ grounded_status.py · introspection_agent.py · tone_analyzer.py · chat_model.py
- runtime/ 33.6k — grounding spine + daemons (93 files)
 - └─ deterministic_grounding_gate.py 4.3k * · grounding_escalation.py
 - └─ evidence_ledger/store/arbitration.py · memory_evidence.py
 - └─ persistence_gate.py · truth_report.py · control_contracts.py
 - └─ response_contracts/packets/policy.py · final_response_assembly/provider.py
 - └─ user_visible_response_surface.py · personal_memory_*.py · reflection.py
 - └─ background_tasks.py · self_improvement.py · code_monitor.py
 - └─ capability_sync.py · pending_proposal.py · runtime_policy.py
- memory/ 8.0k — remember (13 files)
 - └─ memory.py 4.5k Memory · SQLite + FTS5 *god-file
 - └─ vector_store.py FAISS index
- perception/ 9.5k — sense (24 files)
 - └─ vision.py · analyze_image/csv/pdfs/mesh.py · ambient_vision.py
 - └─ audio_stt.py · local_whisper_stt.py · voice_worker(_streaming).py · eli_listen.py
 - └─ tts_router.py · os_controller.py · screen_locator.py · gaze_engine.py
- planning/ 4.2k — proactivity (24 files)
 - └─ proactive_daemon.py · habits_scheduler.py · habits.py · jobqueue_cli.py
- coding/ 2.1k — CodeAgent (plan→search→verify→repair, 12 files)
- learning/ 4.3k — LoRA self-training (14 files)
 - └─ lora_trainer/eval/guard.py · dataset_builder/filters.py
 - └─ bootstrap_phi3_base.py · base_model_resolver.py · training_preflight.py
- plugins/ 5.9k — manager + bundled plugins (34 files)
 - └─ calendar · document_reader · media · notes · pomodoro · weather · system_stats · tts · web · web_automation
- world/ 1.8k — EliWorld (world_event_bus, local_world_bridge)
- core/ 9.4k — infra
 - └─ netguard.py offline failsafe + allow_network()
 - └─ paths.py · portable_paths.py · legacy_paths.py · db_paths.py
 - └─ runtime_settings.py · config.py · grounding.py
 - └─ hardware_profile.py · startup_hardware_optimizer.py · dynamic_runtime_budget.py
 - └─ model_download.py · first_run.py · first_run_wizard.py
- gui/ 26.1k — PySide6 desktop
 - └─ eli_pro_audio_gui_v2_0.py 12.6k *god-file · app.py · labs_tab.py 5.7k

```

├─ panels/ (startup.py: model picker + FirstBootWizard, HardwareTuningDock)
├─ tools/ 7.5k — image_engine · news · document tools
├─ contracts/ 0.7k · cli/ 0.1k · system/ 0.3k · utils/ 0.9k

```

D. On-disk data layout

```

artifacts/
├─ db/
│   ├── user.sqlite3  memories(+_fts) · conversation_turns · conversations ·
│   │                 kg_entities(+_fts) · kg_relations · recall_log ·
│   │                 runtime_events · news_articles(+_fts) · news_reflections ·
│   │                 habits/habit_events/habit_rules · observations ·
│   │                 learning_replay · working_memory_pins · user_patterns ·
│   │                 session_summaries · corrections · failures
│   └─ agent.sqlite3  agent_dispatches · agent_metrics · improvements ·
│                     failures · code_patches · error_tracking · observations
├─ vectors/index.faiss  semantic index (embedder: nomic-embed-...Q4_K_M.gguf)
├─ conversations/ + archive/
├─ runtime/users/<uuid>/user_profile...  per-user profile
├─ runtime_snapshot.json  live model/runtime truth
├─ world/{snapshots,ledger,journal}/ · proactive/ · incidents/<date>.jsonl
├─ documents/ · scripts/ · analyze_image_*/
├─ config/  settings.json (gitignored) · templates/settings.template.json ·
│           settings.example.json · plugins_state.json
├─ models/  <your>.gguf · embeddings/ · whisper/ · image/  voices/ *.onnx (Piper)

```

E. The flow in one line

```

INPUT → ROUTER → ENGINE └─ FAST-PATH → OUTPUT (verbatim)
                        └─ ORCHESTRATOR (all CHAT modes, gradient)
                            ├── retrieve_for_turn() → MEMORY
                            ├── dispatch_specialists() → 15-agent BUS
                            └─ broker.infer() → finalize_turn() ($12)
GATES: netguard · persistence · grounding · escalation · governor
INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler

Companion docs: architecture.md (prose, every subsystem) · diagrams.md (pipeline / memory
/ gating close-ups) · tools/eval/README.md (the measurement layer). “”

```